MACHINE LEARNING

WHAT IS DATA?

RAW FACTS & OBSERVATIONS

Raw facts or observations.

DIVERSE FORMS

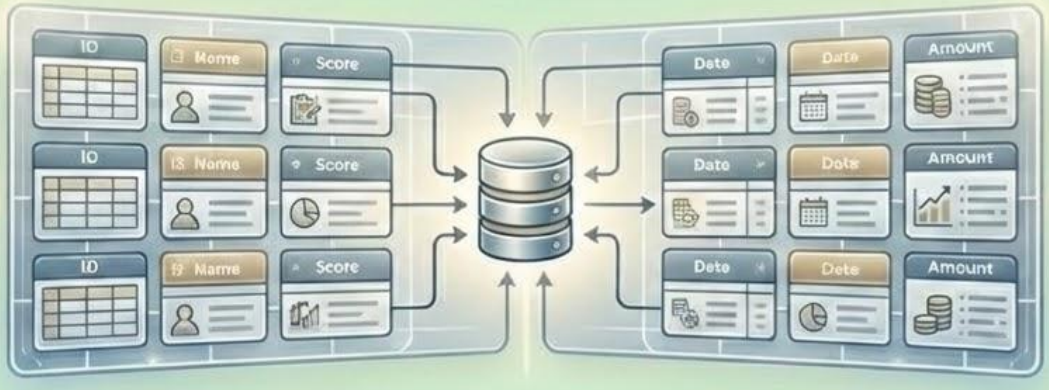
It can be numbers, text, images, audio, video.

NEEDS CONTEXT FOR MEANING

Data alone has no meaning.

TYPES OF DATA

STRUCTURED DATA



Organized in rows and columns.
Stored in tables or databases.
Easy to analyze.

Example: Student marks sheet, bank records.

2.Unstructured Data

Not organized in a fixed format

Does not follow rows and columns

Harder to analyze

Example: Images, videos, audio, social media posts

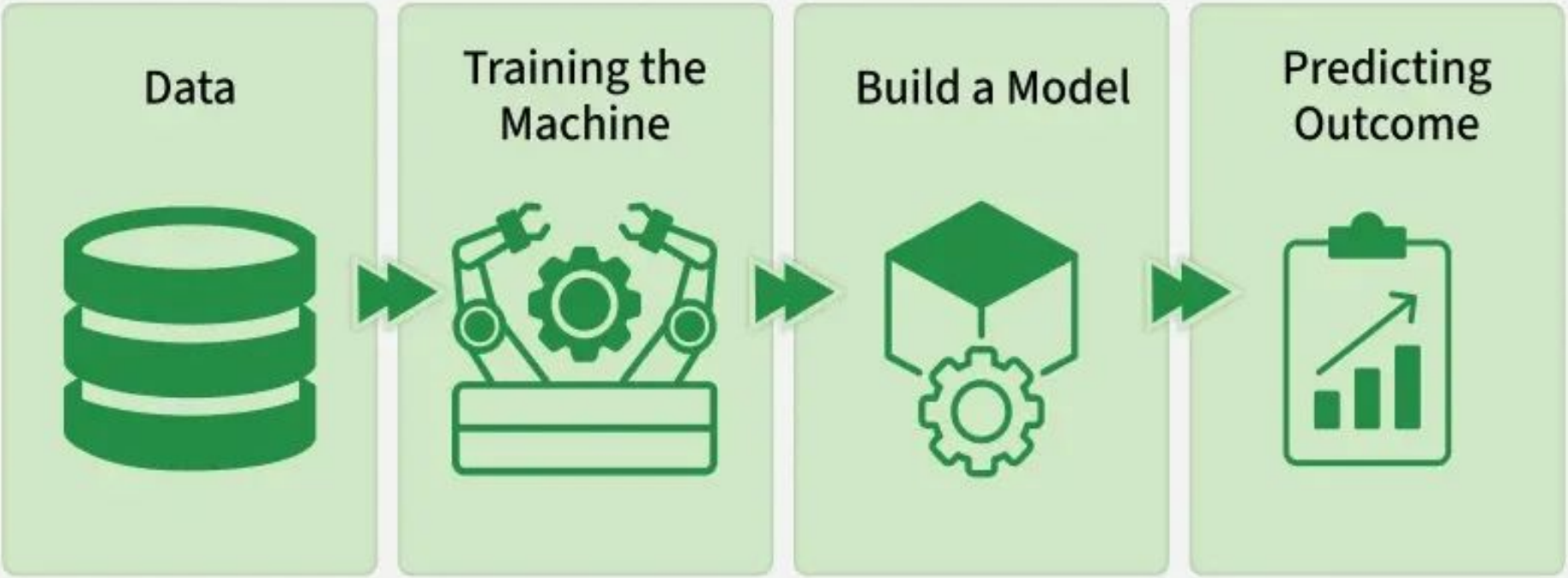
What is Machine Learning?

- Machine Learning is a branch of Artificial Intelligence
- It enables computers to learn from data
- The system improves its performance through experience
- No need to explicitly program every rule

Example:

- Spam email detection
- Movie recommendations
- Face recognition

Introduction To Machine Learning



Types of Machine Learning

Supervised Learning

Learns from labeled data.

Unsupervised Learning

Finds patterns in data without labels.

Reinforcement Learning

Learns through trial and error.

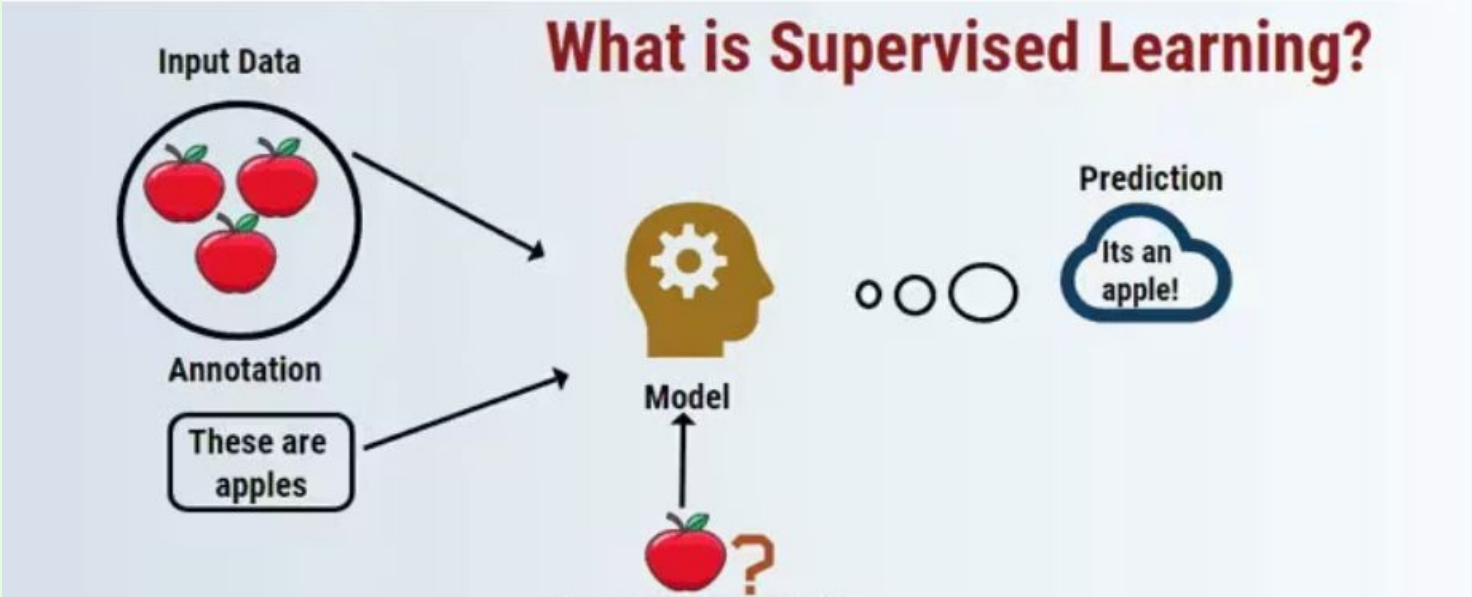
Supervised Learning

- Supervised Learning is a type of Machine Learning
- The model learns from labeled data
- Each input has a correct output (label)
- The goal is to learn the relationship between input and output

Examples:

- Email spam detection
- House price prediction
- Disease prediction
- Face recognition

Basic Concept



Types of Supervised Learning

1. Regression

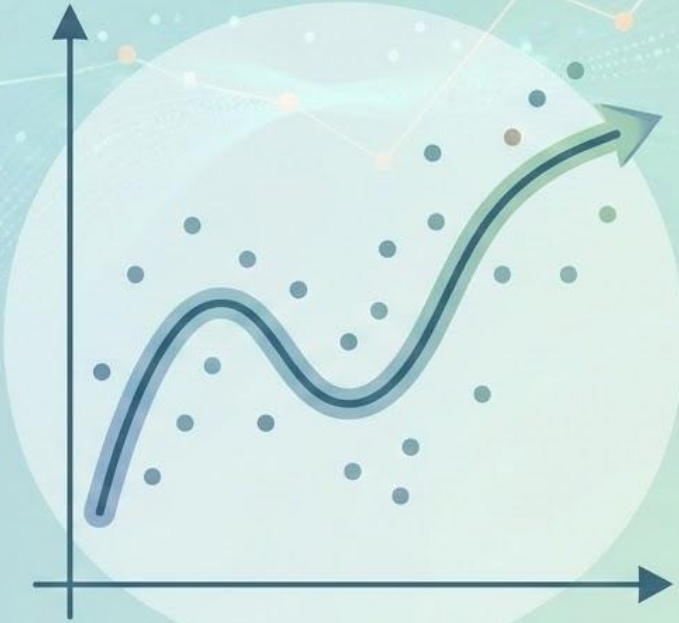
Regression is a supervised learning algorithm used to predict continuous numerical values. The output is a continuous quantity, not a category.

Examples:

Predicting house price

Predicting temperature

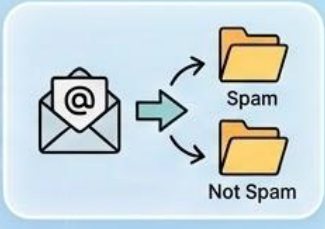
Predicting sales revenue



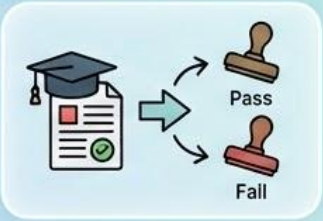
Types of Supervised Learning

2. Classification

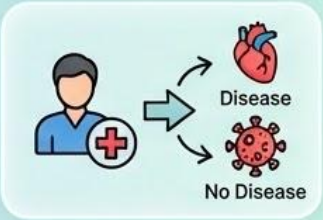
Classification is a type of supervised learning used to **predict categories** or **classes**. The **output** is a **label** instead of a number.



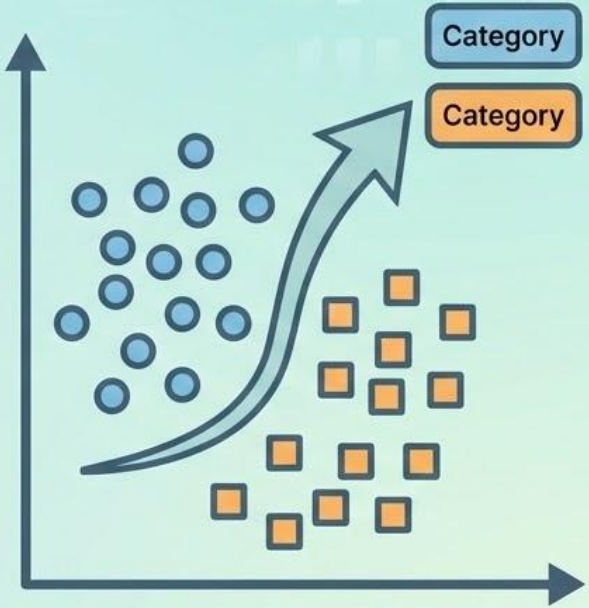
Email
(Spam or Not Spam)

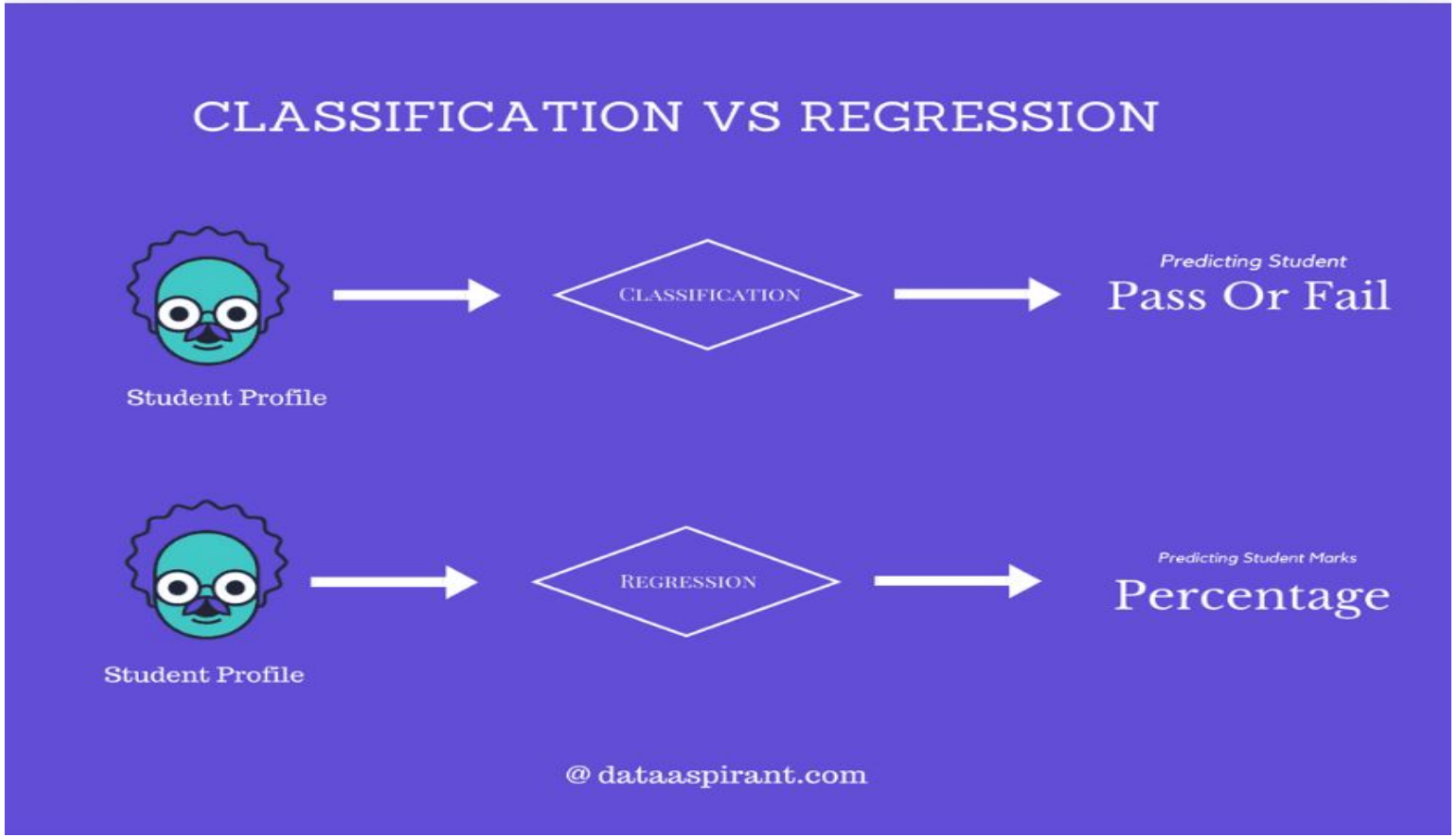


Student Result
(Pass or Fail)



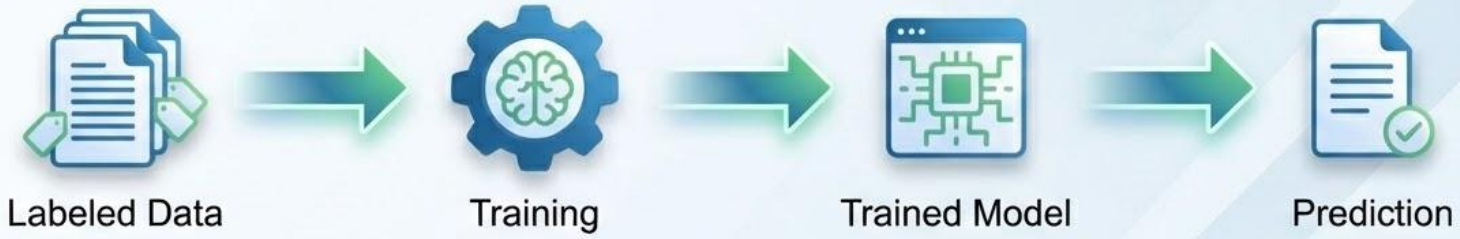
Medical Diagnosis
(Disease / No Disease)




CLASSIFICATION VS REGRESSION
Student Profile
CLASSIFICATION
Predicting Student
Pass Or Fail
Student Profile
REGRESSION
Predicting Student Marks
Percentage
@ dataaspirant.com

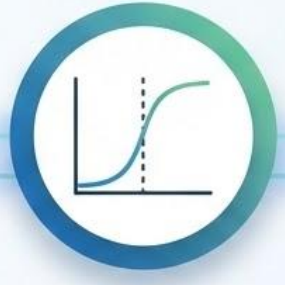
How Supervised Learning Works

1. Collect labeled data
2. Train the model
3. Model learns patterns
4. Model predicts new data

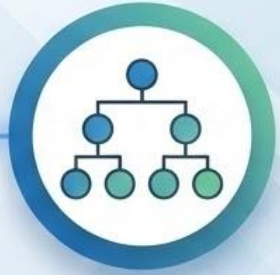


Algorithms Used

Linear Regression



Logistic Regression



Decision Tree

K- Nearest Neighbour

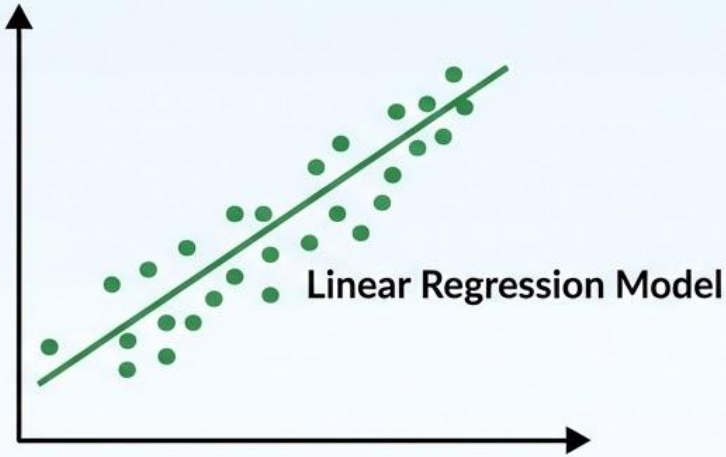
Support Vector Machine

Linear Regression

Used to predict continuous values

Output is a number

Finds a straight that best fits the

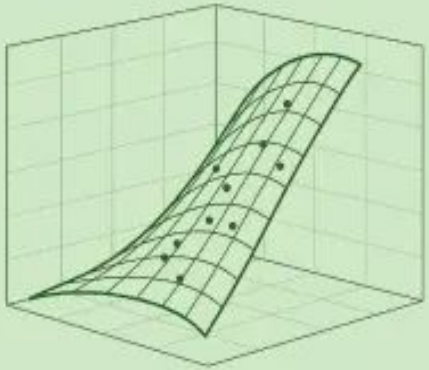


Types of Linear Regression

Simple Linear Regression

Predicts the dependent variable using a single independent variable.

Multiple Linear Regression

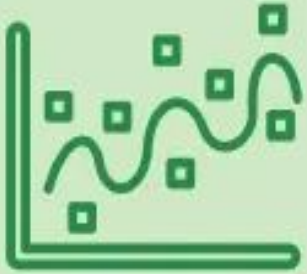


Uses two or more independent variables to predict the dependent variable.

Real-World Use Cases of Linear Regression

Stock Market Prediction

Real Estate Price Prediction



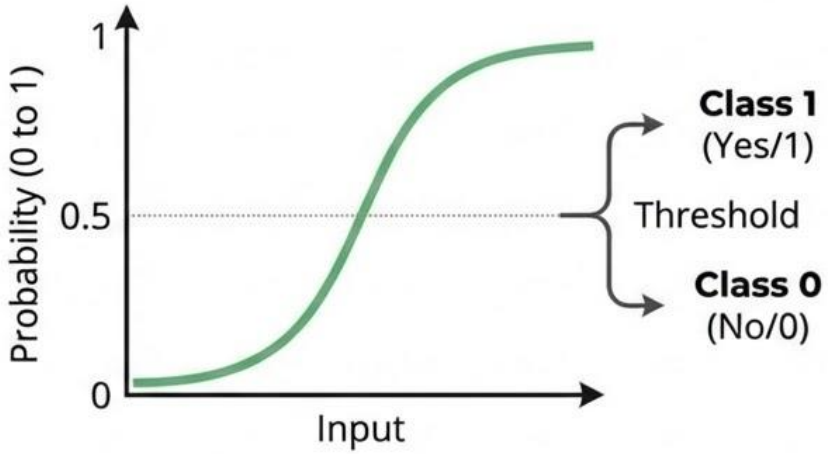
Medical Risk Prediction

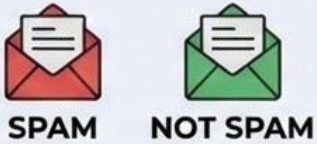
Sales Forecasting

Logistic Regression

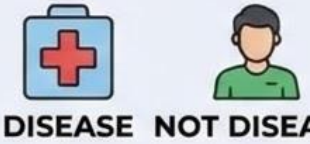
Key Concepts

- Used for classification problems (e.g., Yes/No).
- Predicts the probability of an event occurring.
- Output is a categorical value (0 or 1).





Spam Detection



Disease Diagnosis

Types of Logistic Regression

Binomial

Two classes (0 or 1)

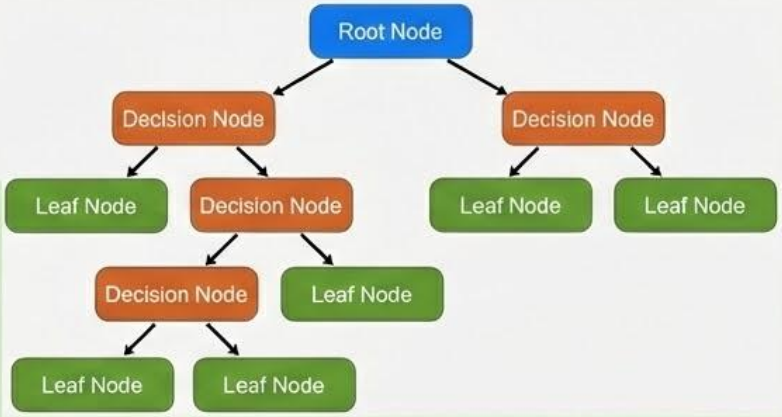
Multinomial

More than two unordered classes (cat, dog, sheep)

Ordinal

More than two ordered classes (low, medium, high)

Decision Tree

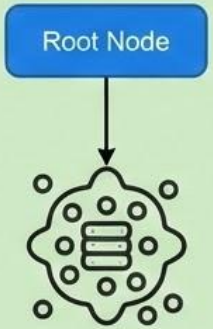


- Decision Tree is a supervised learning algorithm
- It is used for **classification** and **prediction**
- It works like a **flowchart** with questions and **answers**
- Each decision leads to another branch until a final result is reached.

Components of a Decision Tree

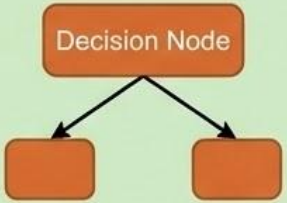
1. Root Node

- The starting point of the decision tree
- Represents the entire dataset



2. Decision Node

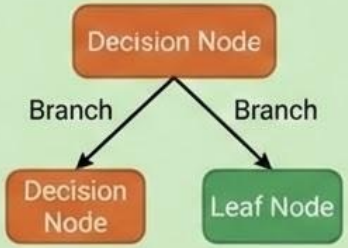
- A node where a condition or decision is applied
- Splits the data into different branches



Components of a Decision Tree

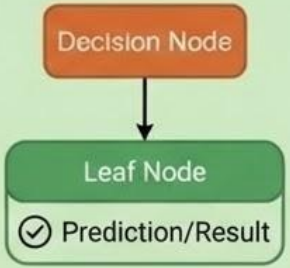
3. Branch

- Represents the outcome of a decision
- Connects one node to another

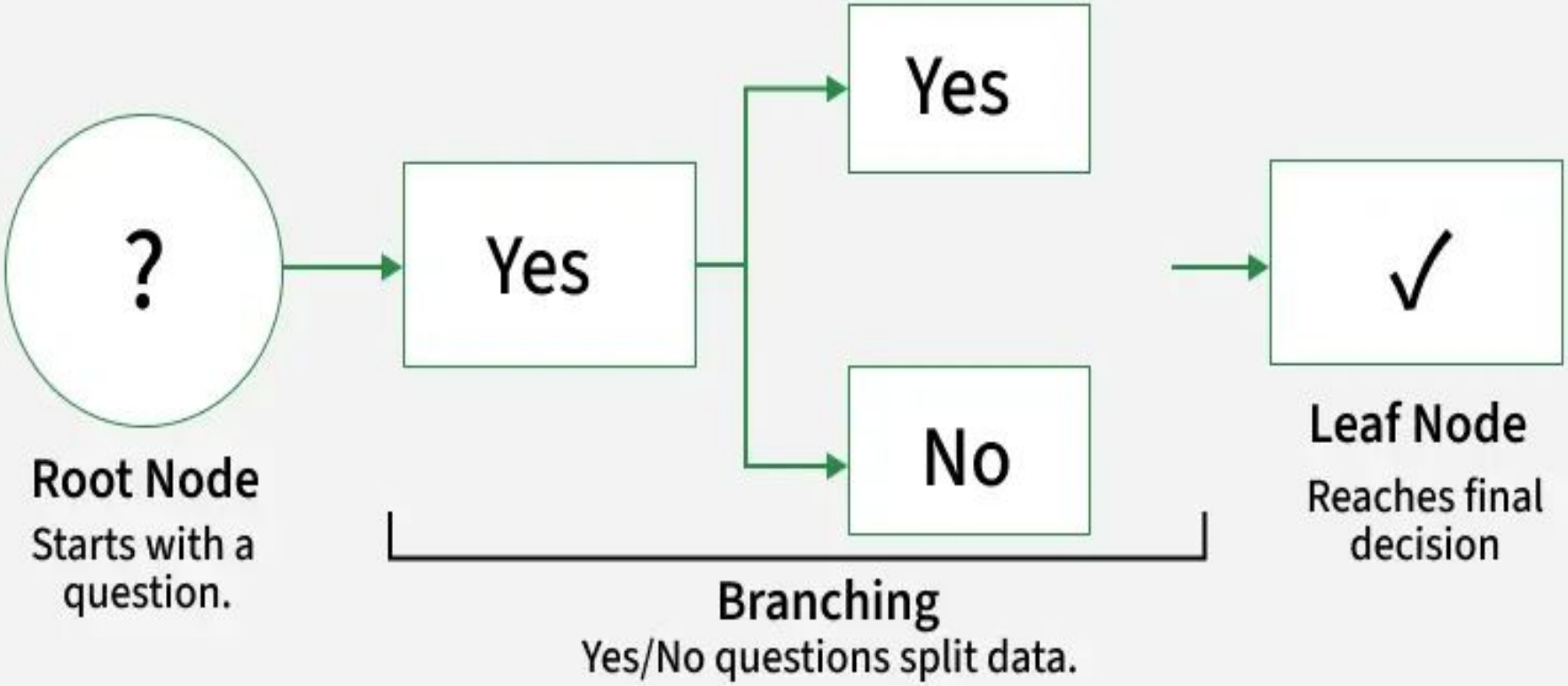


4. Leaf Node

- The final node of the tree
- Represents the final prediction or result



How Decision Trees Work?



Applications of Decision Trees

01 Loan Approval

02 Medical Diagnosis

03 Predicting Exam Results

04 Marketing Campaigns

05 Fraud Detection

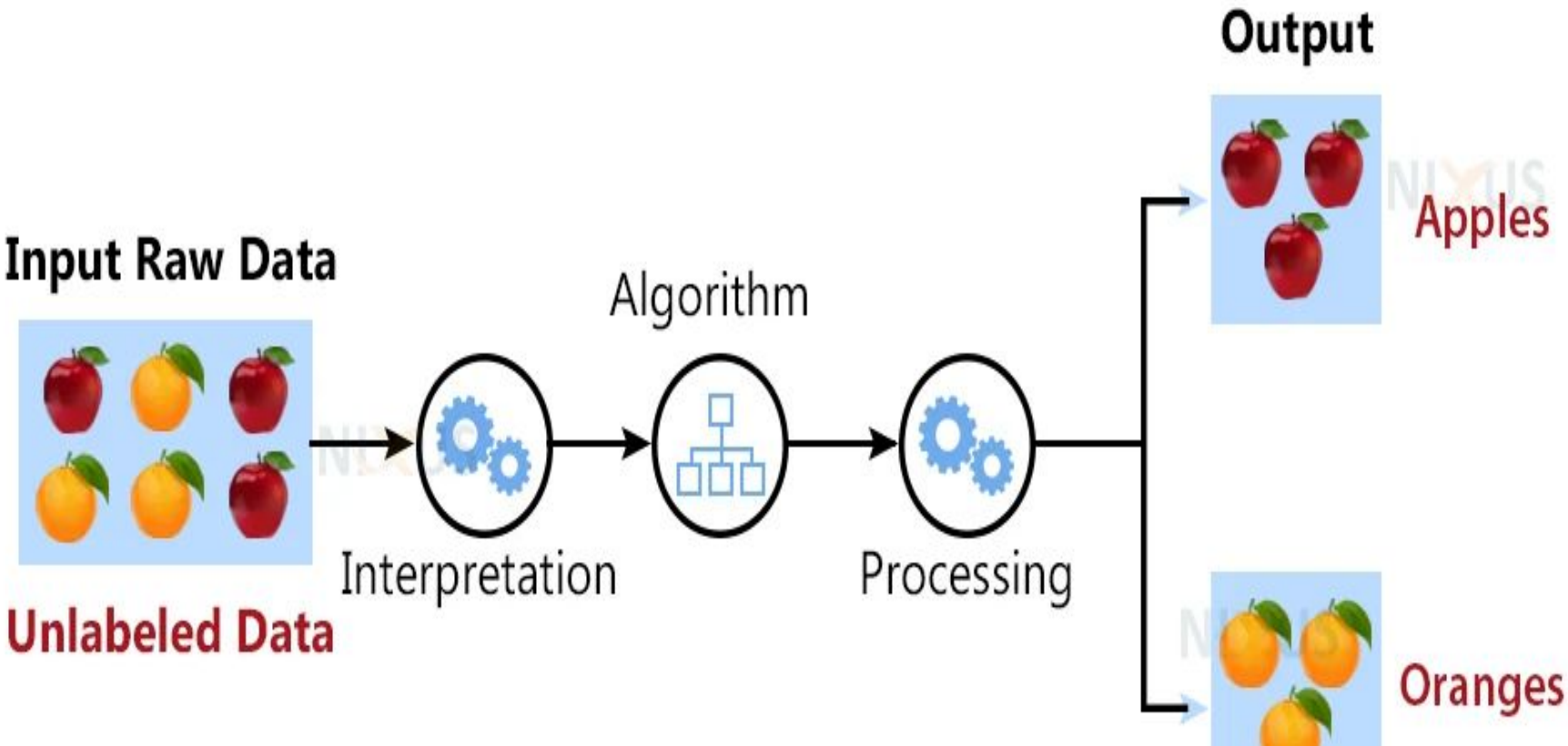
06 Customer Churn Prediction

What is Unsupervised Learning

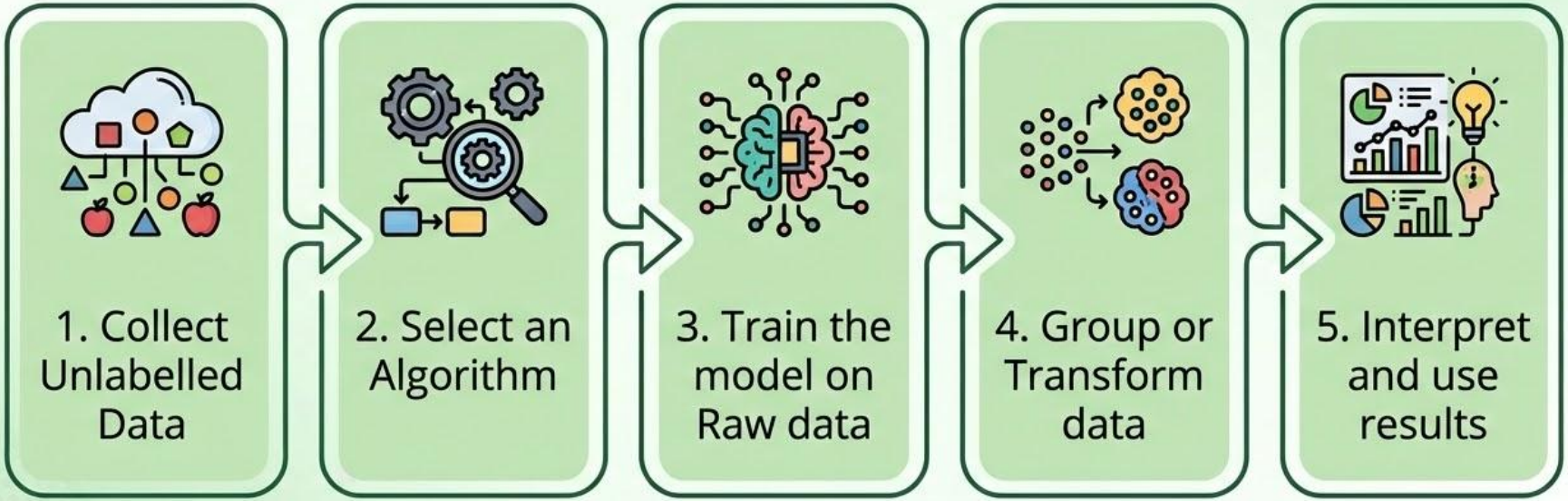
- Unsupervised learning is a type of machine learning
- The model learns from unlabeled data
- There is no correct output provided
- The goal is to find hidden patterns or structures in data

Discovering Hidden Structures

Unsupervised Machine Learning



How Unsupervised Learning Works



Algorithms Used:

K-Means Clustering

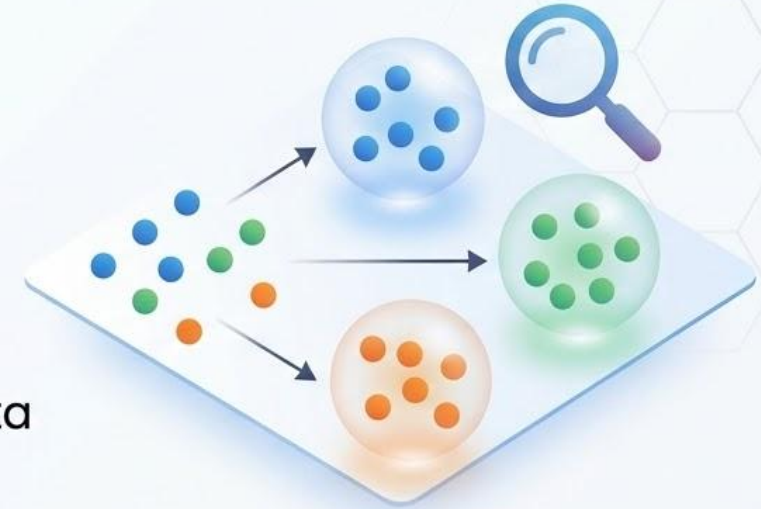
Principal Component Analysis (PCA)

Apriori Algorithm

Association Rule Learning

Clustering

- Clustering is an **unsupervised learning** technique
- It groups **similar data points** together
- Data points in the same cluster share similar characteristics
- It helps to discover **hidden patterns** in data



Examples:

- Customer segmentation
- Grouping similar documents
- Image segmentation
- Market research analysis

Applications

Customer segmentation in marketing

Grouping similar news articles

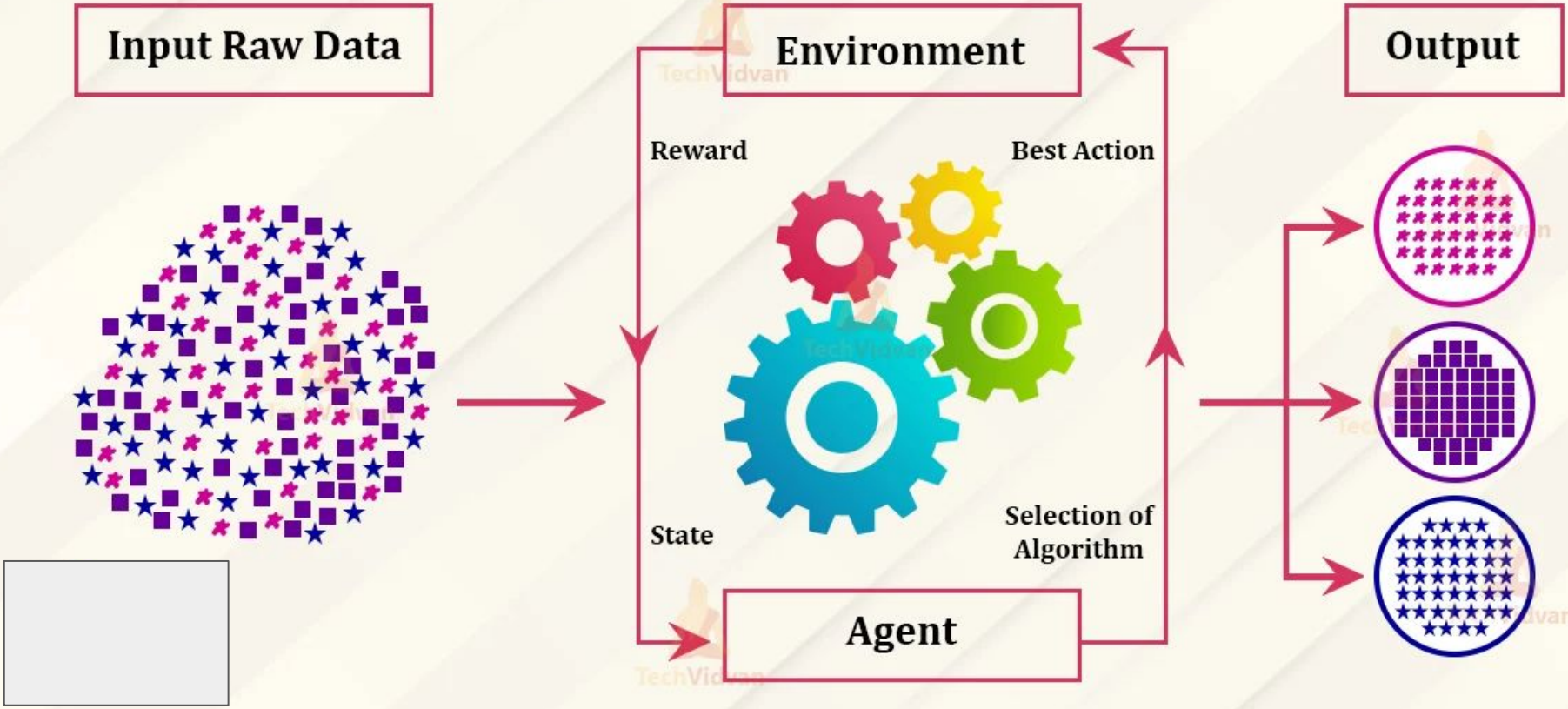
Recommendation systems (movies, products)

Market basket analysis (items frequently bought together)

Reinforcement Learning

- Reinforcement Learning is a type of **machine learning**
- The model learns by **interacting with the environment**
- It improves performance using **rewards and penalties**
- The goal is to **maximize the total reward** over time

Reinforcement Learning in ML



Applications of Reinforcement Learning

Game playing AI

Robotics and automation

Self-driving cars

Recommendation systems

Supervised v/s Unsupervised v/s Reinforcement Learning

Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Data Labeling	Requires labeled data	No labels required	Uses rewards and penalties
Objective	Predict outputs	Identify patterns	Optimize decision-making
Common Algorithms	Regression, SVMs, Neural Networks	Clustering, PCA, Autoencoders	Q-Learning, DQN, Policy Gradient
Applications	Healthcare, Finance, Marketing	Cybersecurity, Retail, NLD	Robotics, Gaming, Finance